

SPECTRAVIDEO USERS GROUP -- Wgtn

Newsletter Number 11 -- June 1985

The next meeting is on Monday, July 22th at 7.30 in the Staff Common Room, Wellington Clinical School, Main Street Newtown.

ON JUNE 22 WE WILL HAVE A FAMILY EVENING AT THE STUDENT UNION RESTAURANT, 2nd FLOOR, STUDENT UNION BUILDING, VICTORIA UNIVERSITY. FOR CATERING PURPOSES WE MUST KNOW NUMBERS WHO CAN ATTEND ASAP. PRICES WILL BE AS FOLLOWS ...

\$14.00 PER ADULT

\$ 8.00 PER CHILD UNDER 12

\$35.00 FOR TWO ADULTS & TWO OR MORE CHILDREN

THIS WILL RUN FROM 5.00 IN THE AFTERNOON THROUGH TO 9.30 OR THEREABOUTS, AND WILL INCLUDE A SMORGASBORD DINNER, COMPETITIONS WITH PRIZES SUCH AS PRINTER PAPER, FLOPPY DISKS, ETC FOR THE KIDS AND NOT SO YOUNG AS WELL, DEMONSTRATIONS OF SOFTWARE, HOPEFULLY A MOVIE ABOUT BONDWELL/SPECTRAVIDEO. THIS IS SOMETHING FOR ALL THE FAMILY AND A CHANCE FOR YOUR CHILDREN TO SHOW OFF THEIR GAMES ABILITY IN THE COMPETITIONS. PLEASE FILL IN THE FORM ON THE CENTRE PAGE AND RETURN TO P.O BOX 7057 ASAP WITH PAYMENT TO ATTEND. TRY TO GET REPLIES BY JUNE 15 PLEASE.

Wellington SV Users Group

Secretary / Newsletter Editor

Don Stanley Ph 896379 (hm), ~~746906~~ (work)

Treasurer

Colin Chin Ph 325656 (hm)

Committee members

Garry Clark 367872

Ross Wakelin 793948

Dennis Timmons 837200

Darryl Alison 85522 (Paraparumu) (hm) 399510 (wk)

Specific areas of responsibility for committee members are ...
Don -- secretary & newsletter ; Colin -- Treasurer & MSX develop-
ments ; Garry -- Radio Access & public domain tape software ;
Ross -- National Committee ; Dennis -- library ; Darryl -- prog-

Laurie Callender

Richmond Road

RD 3

New Plymouth

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IMPORTANT NOTICE

The JULY meeting will be our AGM . Please come along to this as there will be several items of club policy to vote on which will affect club organisation in the future. Club members will receive an agenda with the JULY newsletter . Anybody who wishes to have some subject on the agenda for this please let me know asap . Also any nominations for committee positions are being accepted now.

And we are starting a juniors page(s) in the magazine . Come on kids , any questions about the computer , anythings you want to comment about write to us and we will answer it through our juniors page .

Congratulations

To JAMES WILKINSON, Aged 9
from Lower Hutt who has won an SV 728 MSX computer at the PC Show in Auckland. James demonstrated this machine at our last meeting , it certainly looks interesting (review in latest Bits & Bytes).

Nice one James

New members ...

Welcome to ... John Hodgson , Todd Davies , Ian Montgomery , Mr P Freeman , Ted Andrews , Pat & Chris Zalewski , Laurie Callender , Deane Whitmore , James Wilkinson

Note that expiry date for subscriptions is printed on the address label of newsletter.

Number of Members = 146

Newsletter Contributions

If you are submitting programs to go in the newsletter they MUST satisfy the following conditions or they will NOT be printed.

- 1 ... be typed/printed on A4 paper with about 5 lines left blank at top and bottom of page and an inch at least around the sides
- 2 ... typed with a dark ribbon OR printed using emphasized print

mode

3 ... include a description of what the program does , typed on A4 paper with example input

4 ... In the case of programs above 20 lines in length they should be sent on cassette , in this case there is no need for the printout as I can merge them into the newsletter. Enclose a stamped addressed envelope for return of cassette

5 ... line numbers should be continuous -- do a RENUM before printing the programs

6 ... PLEASE MAKE SURE THEY WORK . and would readers please note that I am NOT responsible for programs sent by people which do not work .

7 ... In the case of screen dumps from 3d plotters please indicate the function used and try to get the print as dark as possible -- not always easy i know .

All other newsletter contributions should be sent to me at P.O Box 7057 , Wellington South. They may miss being printed for a month if sent to any other box or given to anyone else. Articles should follow 1 and 2 above.

Deadline for Newsletter articles is the 30/31st of each month.

Discounts for members .. All discounts offered by Einstein Scientific and Microstyle Computers do NOT include hardware. Discounts may be offered on such peripherals as floppy disks, cassettes , books , furniture, printers , monitors but NOT on SpectraVideo hardware. Individual dealers may not offer all the goods above at discounted rates , inquire from the shops just what discounts are available to club members. Note that to obtain a discount on any goods you must produce your current membership card.

Software Competition

A copy of Sector Alpha , plus 2 year subscriptions to both the user group and Bits & Bytes has been won by Deane Whitmore for his program FUTURISTIC VOYAGE , an educational program which teaches you about the solar system as you visit various planets.

Second prize of a \$50 purchase from Einstein has been won by Iain Scott for his program testing knowledge of chemical symbols , also included in this prize are a 1 year sub to both the user group and Bits & Bytes .

Thanks to all those who entered , and also to both MICROSTYLE &

EINSTEIN for donating prizes. The judging was carried out by staff of RADIO ACCESS , thanks to them also .

GAMES SCORES

Spectron	11710	Robert Campin
Sector Alpha	666400	Gavin Knight
Kung Fu	38170	Taane Clark
	17690	Garry Clark (*)
S C Force	748600	Deirdre Chin
Telebunny	87810	Colin Chin
	26470	Don Stanley (+)
Omac	35360	Richard Bremford
	25700	Don Stanley (+)
Ninja	65200	Richard Bremford
Tetra	266600	Nicola Tuinman
Turboat	68390	Garth Thornton (*)
Sasa	118240	John Tuinman
Frantic Fred	11900	Don Stanley (level4)
Flipper Slip	29990	Karen Stanley (level1)

(+) = started in normal level

(*) - started in arcade level

otherwise practice level or unknown

Could the person who got a higher score on TETRA please write back and include THEIR NAME !!!

Extra Club Meetings

The first meeting of our Kapiti subgroup will be on JULY 21st. Please ring Darryl (phone number on page 1) to find out where and time also .

Also attempts are underway to form a Palmerston North sub-group of our group . If you are interested contact John , ph 722 Shannon .

Club Sponsors

Einstein Scientific -- 177 Willis Street , Wellington

MicroStyle Computers -- 198 Jackson Street , Petone

BUGS BUGS BUGS

This problem has been pointed just recently and will affect users who have recently bought their system with a cassette .

In the cassette manual it is stated that you can draw a graphics screen and then save that screen on tape by pressing control/stop and then typing CSAVE "NAME",S . This is not correct. The correct procedure is as follows .

Draw the graphics in your program and at the end of the picture include a CSAVE "NAME",S command . Have the tape RECORD and PLAY keys pressed first and try this

```
10 MOTOR OFF
20 SCREEN 1
30 FOR I=2 to 7
40   CIRCLE (128,96) , 20*I , 10
50 NEXT
60 MOTOR ON
70 CSAVE "SCREEN",S
80 GOTO 80
```

To load it again just use CLOAD "SCREEN"

Public Domain Software

NOTE: A list of software is on page 12.

- 1) Do NOT send us tapes. We will supply a standard tape which will get round the problem of people sending tapes which are too short.
- 2) The cost of tapes will be \$3.00 each and we will also have a charge of \$2.50 to cover handling and return postage. This is charged for each separate lot of copying done , not per tape.
- 3) Any of the PD Software may also be broadcast on Radio Access in our 5 minute slot on the first Saturday of each month.
- 4) The tapes will contain combinations of cassette & disk programs , some may be set up for 80 columns. Adapt them as you wish to suit yourself.
- 5) Send orders with payment to PDS ,P.O Box 7057 , Wgtn South, Wellington
- 6) Send us your public domain software for inclusion in the software library.

And we now have 28 CP/M disks full of public domain software. But be warned , a lot of this require a detailed knowledge of the CP/M bios and changing of such in order to utilise them . Some of the disks appear incomplete eg PASCAL (UCSD version) . PASCAL in particular requires a lot of modification to the BIOS and also requires disks formatted in a special way as it uses its own disk environment . Complete list (with number of disks in brackets) is

wordprocessor (2) help (2) othello (1) stoic (2) BDS-C (2)
ratfor/actor (1) disassembler (1) Z80 assembler (1) pilot
interpreter (1) Cobol-80 compiler (1) Fortran-80 (1) forth
interpreter (1) pascal (1) pascal utilities (3) adventure (2)
algol-m (2) cp/m utilities (2) stage-2 (1)

No , I do not know what half of them are and we would prefer NOT to have people send us disks as they take ages to copy . I suggest that you ask your friendly dealer to either copy what you want or let you make a copy yourself in the shop .

Some of this software is very old and much of appears to be very early versions of software still on the market eg Z80 assembler 1977 revision , Fortran-80 1980 etc .

We are presently negotiating with the local Osborne group to get some more PDS for CPM from the States .

THANKS

Theres a lot of people who work behind the scenes to help get the magazine ready each month. Heres some acknowledgement for the printers at broadcasting who are so prompt with printing when they get the newsletter , to Karen who helps me label them and staple them , to Dennis who posts them out from the hospital and Garry who races around town collecting them and returning them to me. Thanks folks .

And a word of thanks to these people who let our committee turn their residences upside down at committee meetings. Thanks again.

Surveys

The survey forms are pouring in and make interesting reading . I will answer some of the questions in these here

----> the newsletters need more hints for beginners ...

agreed , but if people refuse to contribute anything to the magazine then it is unlikely that columns especially for the beginner will appear . As it stands at present I write 90% of every newsletter and consider that if people won't come forward with articles for beginners then that is one phase of the club

which will suffer . I would like to see a relative beginner writing about their experiences for the newsletter but if no-one is prepared ...

-----> keep a balance of information in the newsletter between disk and cassette information

again , the onus is on cassette users to send in information . My use of the system is 99.9% disk based and the newsletter reflects this

-----> have more contributions from authors with different interests and more programs and utilities

enough said where have all the writers gone

-----> have more tutorials in the magazine on BASIC commands with program examples

this is something I was doing early on in the newsletters with the hope that people would send in articles on some of the newer commands . I never got any at all. Come on you people out there , its your user group so why sit back and take all the time . I spend enough time already on this newsletter , and if no-one else is going to send in contributions it will continue to reflect what I'm doing with my machine

-----> how about a monthly raffle at meetings with maybe a tape program as prize at something like \$1.00 a ticket

sounds great , we'll try this out at our JULY meeting and see how it goes

-----> more tutorials at meetings and better prepared

agreed , I started these after many requests but I haven't prepared them in advance , how about a few volunteers to give tutorials on sprites , sound and so on.

-----> publish a list of club members so members can communicate with each other

this has been suggested before and as soon as someone can come up with a way of doing this without inviting the possibility of such a list falling into the wrong hands we will do it .

-----> buy sell or swap columns in the newsletter

done , beginning this newsletter

-----> needs more information in the newsletter about machine code

this has started now and will continue

-----> more info needed in magazines on direct accessing rom routines and how to use rom routines

slowly and painfully being compiled at present

-----> the WBTN User Group puts out a very good newsletter

which could be even better if a few more people would contribute !

-----> more real life applications and uses in the newsletter

we have members working on Videotext interfaces , MIDI interfaces , incorporating super capacity disk drives and all sorts of other things . If they want to keep them entirely to themselves theres nothing I can do about it

-----> less one up manship in the newsletter and meetings

really I take this very personally . I started this group to share information and learn from others . Everything that I find out about the system goes into the newsletter . If thats going to be considered one up manship then as far as I'm concerned I might as well not bother .

-----> raise the newsletter standard

99% of all comments I receive about the newsletter are very complimentary . This is the other 1% . And from an Auckland reader !

-----> print authors name below each article

All articles not written by myself have authors named

-----> The Bits & Bytes column lets us down ... to esoteric and should be diluted with material to attract potential buyers

I tend to agree , but that column suffers from the same problem as this newsletter --- not enough input from people with

different ideas and interests .

-----> the user group should answer letters from out of town members in a reasonable time

??? I try to answer all mail via the newsletter as most requests , questions etc are of interest to all members . Also I often get two or three similar queries in over a period of a few weeks , replying via the newsletter gets to everyone and saves duplication of work for me .

-----> less ads and repeated material in newsletter....

done , this has appeared in several replies so far , so I will repeat things about 3 monthly rather than monthly as at present

These are the sorts of comments I have received so far. Its up to YOU to help with some of them and keep the HIGH STANDARD of this newsletter and users group up . More next month .

The ZORK assistance line ...

you need the assistance of a character found in the dungeon to open the jewel encrusted egg (did I hear you say WHAT jewel encrusted egg ... you need to climb for it)

National Committee Meeting

Over the weekend of 25/26 May the WGTN group hosted SV owners representing clubs nation wide for a national committee meeting. Many topics were discussed and a complete booklet summarising the meeting will be produced for anyone interested -- about a month.

The following motions sum up the meeting and have been adopted by all users groups present at the meeting .

1 .. that a National SV working committee be formed with 1 rep from each group to channel research efforts into particular subgroups covering specific areas .

2 .. that we support a national SV magazine and form a working committee with CDL about this

3 .. to channel all public domain basic system software through the WGTN Users Group from where it is distributed to other user groups , and to channel all CP/M software through the New Plymouth User Group in the same manner

4 .. That the national committee forms a sub-committee to form a ring binder of information with updates being passed through the SV magazine. This ring binder would be made available to anyone at a cost aimed solely at recovering costs.

5 .. the national committee will be known as the NEW ZEALAND SVI USERS ASSOCIATION .

6 .. all persons elected to the NZSVIUA committee should declare any commercial interest in SVI .

7 .. a levy of \$1.00 per financial member of each club will be made to the NZSVIUA to assist in the running of same .

8 .. in order to keep costs as low as possible , national committee meetings will be shared around committee members respective localities

9 .. The first executive committee of the NZSVIUA has been elected as follows ...

Chairman -- Don Stanley (WGTN)

NI Secretary -- Deane Whitmore (N.PLY)

SI Secretary -- Lester Reilly (CHCH) (not yet confirmed)

CDL representative -- Peter De Zwart

This committee will have a first meeting in JULY . All user groups have until first JULY to make submissions regarding the running and membership of the committee . It must be stressed that there is no obligation on the part of users groups to join this committee .

If any member of the WGTN group has any queries or comments about this , including specific ideas on policy , please contact me asap either by mail or phone .

A complete conference proceedings , including comments of interest to all SV owners from CDL is currently being prepared .

Library

The latest SV-Access from Canada and newsletter of the Montreal Users group are now in the library . (The Montreal one is printed in both English & French , but the French bit looks to have a few extra things about the ROMS , does anyone in the club speak French so we can get it translated) . We also have the latest magazine from the Hamilton users group which has details of how to use the other graphics mode not implemented in BASIC.

SPECTRA	VIDEO	RECOMMENDED	RETAIL	PRICE	LIST
1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36
37	38	39	40	41	42
43	44	45	46	47	48
49	50	51	52	53	54
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61	62	63	64	65	66
67	68	69	70	71	72
73	74	75	76	77	78
79	80	81	82	83	84
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337	338	339	340	341	342
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385	386	387	388	389	390
391	392	393	394	395	396
397	398	399	400	401	402
403	404	405	406	407	

MAY 1985

MSX RETAIL PRICES

<u>MSX LIBRARY BOOKS</u>		SVI-728	SPECTRAVIDEO MSX COMPUTER	\$ 695.00
SYSTEME	MSX EXPOSED	\$ 39.95	MSX DISK DRIVE	\$ 995.00
SYSTEMG	MSX GAMES	\$ 39.95		
SYSTEMI	MSX INTRO. PACK (INC. CASSETTE)	\$ 49.95	MSX 80 COLUMN CARD	\$ 375.00
SYSTEMC	COMPLETE MSX PROGRAMMERS GUIDE	\$ 79.95	MSX 64K RAM	\$ 345.00
SYSTEMS	STARTING WITH MSX	\$ 19.95		
SYSTEMS	BEHIND THE SCREENS OF MSX	\$ 29.95	MSX RS232	\$ 245.00
SYSTEMC	MACHINE CODE ON THE MSX	\$ 29.95	MSX DATA CASSETTE	\$ 89.00
SYSTEMG	GETTING MORE FROM MSX	\$ 29.95	MSX JOYSTICK II	\$ 29.95
		SVI-208	MSX CASSETTE CABLE	\$ 19.95
		SVI-PARAM	SVI-728 PRINTER CABLE	\$ 49.95

**..... and Spectravideo sets the pace
with the FIRST MSX Personal
Computer in New Zealand.**

**SVI-MSX. A giant leap forward
in Computer Compatibility**

For Disc and Cassette Systems on Cassette

CASSETTE TAPE 1

Title:	Description:
MUSIC	3 note chord on 1 Key
BLITZ	Creatures from the sky
gofh	Gates of Hell Adventure
robons	Swamp creature game
cycle	Motor cycle jumps
tron	Avoid trace Game
Aust	Australian Map
Snake	Snake whose tail grows
Planet	Planet X
hang	Hangman
hand	Hi res screen drawer

CASSETTE TAPE 4

pub 4	Intro
fred	The disk utility
hexlo	M/code Loader/Disass
codeco	Code copy to Ram
drop 1	SV title drops
drop 2	SV title drops(two)
circle	Circle graphics
triang	Triangle graphics
Saturn	Saturn and its rings
disco	Disk copying program
exampl	Basic verses M/C
yatzie	Yatzie Dice game
fuel	Fuel consumption
Shapes	Shapes designer
sports	Sports team draw
blackb	blackboard
don 1	disassembler
budget	Home budgting
h/hous	Haunted House Game
screen	Screen dump Saturn

CASSETTE TAPE 5

Bacter	Circle cmd Graphics
Maths	Speed Maths
AceSht	Ace Shot
Shanes	Shanes Sprites
Mcode	Machine Code
Marque	Large graphics
Symbol	Chemical Symbol Tester
Vernie	Vernier Caliper Tutor
Flight	Flight !!!!

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10 REM  SPRITE DEMO - BY ELLIOTT YOUNG
20 DEFINT A-Z:DIM C(30),N(16):COLOR 11,0,0:R=RND(-TIME)
30 SCREEN0,0:INPUT"# of colors (ENTER for all):";N:IF N=0 THEN 40 ELSEFOR I=1 TO N
:INPUT"number";N(I):NEXT:GOTO50
40 FOR I=0 TO 14:C(I)=I+1:NEXT:C(15)=RND(5)*15+1:C(16)=RND(5)*15+1:N=17:SCREEN1
,3:GOTO 70
50 V=17/N:G=1:SCREEN1,3
60 FOR I=0 TO 16 STEP V+1:FOR X=0 TO V:C(I+X)=N(G):NEXT:G=G+1:NEXT
70 FOR I=82 TO 146 STEP 4:LINE(115,96)-(I,33):LINE(115,96)-(I,161):NEXT:FOR I=33 TO
161 STEP 5:LINE(115,96)-(82,D):LINE(115,96)-(146,D):NEXT
80 A$="SPECTRAVIDEO":FOR I=0 TO 11:LOCATE 200,I*15+4:X$=MID$(A$,I+1,1):COLOR
I+2:PRINTX$:LOCATE 200,5+I*15:PRINTX$:LOCATE 20,4+I*15:PRINTX$:LOCATE 20,I*15
+5:PRINTX$:NEXT
90 FOR I=1 TO32:B$=B$+CHR$(255):NEXT:FOR I=0 TO 16:SPRITE$(I)=B$:NEXT
100 FOR I=0 TO 3:PUTSPRITEI,(50+I*32,1),C(I):NEXT:FOR I=3 TO 0 STEP-1:PUTSPRITEI+8
,(50+(3-I*32,161),C(I+8):NEXT
110 FOR I=0 TO 3:PUTSPRITEI+12,(50,33+(3-I*32),C(I+12):PUTSPRITEI+4,(146,33+I*32),
C(I+4):NEXT
120 FOR I=0 TO 16:VPOKE 6915+(I*4),C(I):NEXT:FOR I=0 TO 15:C(D)=C(I+1):NEXT:C(16)=
C(0):IF NOT STRIG(0) THEN GOTO 120
130 COLOR 11,0,0

```

For Sale

- 1 ... Super Expander , 1 Disc Drive and controller , CPM & Software , \$900
- 2 ... 1 Disc Drive and software , \$600
- 3 ... Centronics interface , \$150

Laurie Callender , New Plymouth , Ph Lepperton 760 or contact Don Stanley for address .

DISK BASIC FILE BLOCK

Of interest to disk users will be this information about the OPEN statement .

When you OPEN a file , a portion of RAM is used to keep various attributes of the file in . This area is found by using the statement PRINT VARPTR(#1) or whatever file number you are interested in .

One byte which particularly interesting in this file block occurs at VARPTR(#1)+7 . The possible values are ...

filetype ---	BINARY	ASCII	MCODE	SCREEN
--------------	--------	-------	-------	--------

Attribute

NONE	82h	02h	03h	A2h
R <i>random?</i>	C2h	42h	43h	E2h
P <i>protected</i>	92h	12h	13h	B2h

For a new file , ie not already on the disk , this byte becomes 02h , ie all files opened for the first time are assumed ASCII .

According to the DISK BASIC MANUAL there are other values for this as well , these look as though they are used with RANDOM access , I am exploring these at present .

How do we use this , or where do we use this . Suppose you write a program which needs to read from a file and you know the file should be ASCII. All you need is .

```
1000 OPEN "1:infile" FOR INPUT AS 1
1010 IF VAL (HEX$(PEEK (VARPTR(#1)+7))) <> &H02 THEN 5000
1020
```

1030 normal processing
1040

.....

5000 PRINT "INPUT FILE NOT ASCII / OR IT IS PROTECTED"
5010 CLOSE : END

The other bytes in this file block are outlined in the DISK BASIC MANUAL and include allocation tables pointers, buffer sizes, last sector accessed and so on. I am exploring these at present, but the above one looks the most useful. This information is also stored on the disk, but this is a much much better way to check the file than that which I presented in newsletter #2.

Little Things

Basic reserves memory locations (hex) F54A/4B for the location which is the beginning of your basic program. F7EE/EF points to the beginning of BASICS variable table. This contains all variables met by the program so far with values and types. This variable table usually occurs 2 bytes after the end of a BASIC program.

The MSX adapter adapts 'most' MSX cartridges. It includes a standard MSX cassette interface and two joystick ports. Evidently it is effectively a 16k MSX machine plugged into the back of the 318/328.

The ROM routines at (hex) ~~3B86~~^{2B86} and ~~3B9F~~^{3B9F} respectively erase and display the function keys. In a machine code program to erase the keys use

CALL ~~3B86~~ 3B86

and to display them use

CALL ~~3B9F~~ 3B9F

And an alternative way to clear the keyboard buffer is DEF
USR=&H7467 : A=USR(0)
351F

Basic Notes .. High Memory

You may have noticed in the last few newsletters that we have been publishing notes about memory locations up around &HF500 and upwards. These locations (up to &HFE78 on the SV318/328) are

known as high memory , or sometimes the BIOS work area . They are a set of locations which the BASIC operating system uses as temporary storage space .

When a program is entered in your machine , the machine code program that controls the system (the dreaded ROM) uses certain of these locations to store information about the program , things like current line number , where the program starts and ends in memory , where the keyboard buffer is and so on. These are updated every time you enter a character. As well as that updating , 50 times a second the system checks the keyboard and various other things , increments the system timer and so on. In fact that check is done whenever the system is switched on UNLESS you run a mcode program that overrides it .

Even when you are doing nothing at the machine , these high memory locations are being used , the timer is being incremented, the buffer is checked and so on .

What sort of things are in the locations and how do we use them. The sorts of things are the current screen mode , colors , function key status , whether in insert mode or not , the keyboard buffer , the current function keys definition and lots lots more . Some we can change , like the location which contains the bottom line on the screen , try POKE &HFA06,240 for instance , some we can change and have no obvious effect or cause a system crash .

The whole point of these locations is as a filing system for system attributes set by BASIC. Being able to alter them gives a lot of control over BASIC , which is why these are appearing in the newsletter . If you want to experiment with these locations , ensure your program is saved first , and let us know the results as there are still many we haven't worked out . Right Deane ?

And also in there from (hex) FE79 onwards are the hooks . These are escape routes from the ROMS . Many ROM routines , before they do anything , go to a hook location . Usually they return straight away , but the point is that you can use the 3 byte long hook to jump to any location you want including to a mcode routine as an alternative to the ROM routine. Again I am slowly , along with several others country wide , working through the roms to find the hooks to see how they can be manipulated .

By the way , if you're unsure where the ROMS occur , they are from (hex) 0000 to 7FFF .

Shadows Music from Garry

```
10 REM [3] Channels with drums * G Clark
20 CLS: PRINT "How do I love thee - by the Shadows": PRINT
30 SOUND7,220: SOUND6,3: REM TURNS ON DRUMS
40 FOR D=7 TO 14: D$="S"+STR$(D): PRINT "PLAY SOUND SHAPE = S";D:
```


[illegible]

I ♥ SVI™
SPECTRAVIDEO



EINSTEIN SCIENTIFIC LIMITED
EINSTEIN SCIENTIFIC LIMITED

EINSTEIN'S have the extras for you:

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3-D OBJECTS

Here is a program that draws a 'wire image' of an object in 3-D. At present the program is set for drawing a cube, but the shapes that can be drawn are innumerable; I will try to explain how you go about making one.

Draw a 3-D representation of the object, numbering the corners(vertices).

Roughly estimate the co-ordinates in each of the three directions for each point (x=horizontal, y=vertical, z=front/back).

	x	y	z
1.	0	0	0
2.	-0.5	-1	0.5
3.	0	-1	-0.5
4.	0.5	-1	0.5

Find a way of drawing the shape in terms of the numbered points:

1-2-3-4-1 and 2-4 (seven vectors in total).

When starting a new line, use a negative number to let the program know.

Therefore, the data for this shape is four vertices with coordinates 0,0,0, -0.5,-1,0.5, 0,-1,-0.5, 0.5,-1,0.5

The seven vectors: -1,2,3,4,1,-2,4

Line 490 contains the number of vertices, line 500 contains the 3-D coordinates(presently set up for drawing a cube), line 510 contains the number of edges(lines), and 520 contains the lines.

The program is only a basis on which to build and modify, incorporate with a screen dump, or allow you to save objects to disk and modify them later, etc.

Use the cursor keys to move the shape around the screen, and N and F for Near and Far respectively. X,Y, and Z do a rotation around that axis, which are currently set to 15 degrees, but can be set to whatever you like by changing line 230 and 240.

One last word - the example shape given here is not very impressive - design your own!!

10 REM 3-D OBJECT VIEWING SYSTEM

BY ELLIOTT YOUNG

20 DEFDBL A-Z:DEFINT I

30 COLOR 11,0,0:SCREEN1

40 PI=3.14159265#

50 Q=0:R=0:S=5

60 READ NV

70 DIM X(NV),Y(NV),Z(NV),XX(NV),YY(NV)

80 FOR I=1 TO NV

90 READ X(I),Y(I),Z(I)

100 NEXT

110 READ NE

120 DIM DR(NE)

130 FOR I=1 TO NE

140 READ DR(I)

150 NEXT

160 GOTO 250

170 A\$=INPUT\$(1)

180 IF A\$="X" OR A\$="Y" OR A\$="Z" THEN GOSUB 370

190 IF A\$=CHR\$(29) THEN Q=Q-.4

200 IF A\$=CHR\$(28) THEN Q=Q+.4

210 IF A\$=CHR\$(31) THEN R=R-.4

220 IF A\$=CHR\$(30) THEN R=R+.4

230 IF A\$="N" THEN S=S-.5

240 IF A\$="F" THEN S=S+.5

250 CLS

260 FL=0

270 FOR I=1 TO NV

280 XX(I)=150*(X(I)+Q)/(Z(I)+S)

290 YY(I)=150*(Y(I)+R)/(Z(I)+S)

300 NEXT

310 FOR I=1 TO NE

320 P=ABS(DR(I))

330 IF DR(I)<0 THEN PSET(XX(P)+140,96-YY(P)):GOTO 350

340 LINE-(XX(P)+140,96-YY(P))

350 NEXT

360 GOTO 170

370 FOR I=1 TO NV

380 A=Y(I):B=Z(I)

390 IF A\$="Y" THEN A=X(I)

400 IF A\$="Z" THEN A=X(I):B=Y(I)

410 A1=(A*COS(PI/12))-(B*SIN(PI/12))

420 B1=(A*SIN(PI/12))+(B*COS(PI/12))

430 A=A1:B=B1

440 IF A\$="X" THEN Y(I)=A:Z(I)=B

450 IF A\$="Y" THEN X(I)=A:Z(I)=B

460 IF A\$="Z" THEN X(I)=A:Y(I)=B

470 NEXT

480 RETURN

490 DATA 8

500 DATA -1,-1,-1,1,-1,-1,1,1,1,1,-1,1,1,-1,1,1,-1,1,1

510 DATA 16

520 DATA -1,2,3,4,5,6,7,4,-7,2,-6,1,8,5,-8,3

530 STOP

This BASIC program patches the CP/M system resident on your disk to change the function key definitions. Because the system itself is changed, these function keys are not lost when the system is rebooted (although other programs may change the key definitions temporarily).

Type the program in (making changes to the key definitions to suit your own requirements), and SAVE on a BASIC disk. Remove the BASIC disk and insert a CP/M disk (use a spare disk to start with, to make sure that the program works). RUN the program. When the Ok prompt reappears, reboot the system (into CP/M) and check the key definitions now in place.

Key definitions may be changed by changing the MID\$ statements. Put your key definition within the quotes "", use a CHR\$(13) if you want the command to be automatically ENTERED, and make up the rest of the string length using the STRING\$ statement. Note that keys 1 and 9 have a total string length of only 11 characters (cf 16 for the others). Characters 12 to 16 of these keys would need to be entered into the next disk sector. (Note that CP/M uses only the first 128 bytes of these disk sectors).

cpm.zap listing.

*I.e. permanently changes CP/M
System Disk.*

```
100 '
110 ' Patches CP/M to set new function keys.
120 '
130 ' by Brian Speirs.
140 '
150 CLEAR6000
160 A$=DSKI$(1,0,2)
170 MID$(A$,118,11)="dir"+CHR$(13)+STRING$(7,0)
180 FIELD#0,128ASB$,127ASD$
190 LSETB$=LEFT$(A$,128)
200 LSETD$=RIGHT$(A$,127)
210 DSKO$1,0,2
220 A1$=DSKI$(1,0,3)
230 MID$(A1$,6,16)="sc"+CHR$(13)+STRING$(13,0)
240 MID$(A1$,22,16)="stat *.*"+CHR$(13)+STRING$(7,0)
250 MID$(A1$,38,16)="ws"+CHR$(13)+STRING$(13,0)
260 MID$(A1$,54,16)="pip"+CHR$(13)+STRING$(12,0)
270 MID$(A1$,70,16)="dir b:"+CHR$(13)+STRING$(9,0)
280 MID$(A1$,86,16)="format"+CHR$(13)+STRING$(9,0)
290 MID$(A1$,102,16)="1copy"+CHR$(13)+STRING$(10,0)
300 MID$(A1$,118,11)="sysgen"+CHR$(13)+STRING$(4,0)
310 FIELD#0,128ASB1$,127ASD1$
320 LSETB1$=LEFT$(A1$,128)
330 LSETD1$=RIGHT$(A1$,127)
340 DSKO$1,0,3
350 A2$=DSKI$(1,0,4)
360 MID$(A2$,6,16)="reboot"+CHR$(13)+STRING$(9,0)
370 FIELD#0,128ASB2$,127ASD2$
380 LSETB2$=LEFT$(A2$,128)
390 LSETD2$=RIGHT$(A2$,127)
400 DSKO$1,0,4
```

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